



Optimizing Multivariate Conditions for Efficient Enzyme-Assisted Polysaccharide Recovery from *Cucurbita moschata* Duch.

Author: Berteanu Tudor Codrut Cristian, Anna Jakubiak-Marcinkowska, Izabela Michalak

Contact: 293772@student.pwr.edu.pl

Abstract

Enzyme-assisted extraction (EAE) is emerging as an innovative and green platform for the recovery of bioactive compounds from plant residues. In this study, polysaccharides were extracted from *Cucurbita moschata* Duch residue pulp using pectinase, and the extraction time, temperature, and pH were optimized using Response surface methodology. Statistical analysis showed that extraction time and pH significantly influenced the polysaccharide yield, while the effect of temperature was less pronounced. A higher yield of polysaccharide was obtained when a longer time of extraction and near-neutral value of pH were used. Also, the respective optimal extraction conditions were 45 minutes of extraction time, 30 °C, and pH 5.07. The obtained results thus confirm the opportunity to conduct valorization of waste materials from pumpkin processing using enzyme-assisted extraction.

Keywords: Enzyme-assisted extraction; Pectinase; Polysaccharides; *Cucurbita moschata* Duch; Response surface methodology;

1.Introduction

Enzyme-assisted extraction, (EAE) is considered a substitute to conventional techniques for extracting valuable compounds using water as a solvent rather than an organic solvent. Enzymes such as pectinase, β -glucanase, β -glucosidase, cellulase and xylanase depolymerize the polysaccharides of cell wall, thereby liberating the linked compound (Aamir et. Al, 2021). This green technique boosts yields from many sources like sweet cherry pomace or sugar beet leaves by hydrolyzing complex matrices, often increasing extraction efficiency by over 40% compared to traditional solvent methods. It's applied in food waste bioremediation to accelerate

decomposition of organics into simpler compounds, supporting sustainable processes like anaerobic digestion.

EAE offers high selectivity, environmental friendliness, and mild conditions that preserve heat-sensitive compounds. However, it requires precise control of pH, temperature, and enzyme choice, with high enzyme costs as a drawback. Ongoing advances in thermostable and reusable enzymes aim to improve scalability. This technique was employed by Umavathi et. al, 2023 for the extraction of polysaccharides from pumpkin (*Cucurbita moschata* Duch). *Cucurbita moschata* (Pumpkin) is a multipurpose species whose fruits and sometimes oil seeds are used for various purposes. It is low in calories and is a significant source of income (Ezin et. al, 2022). Thus, this species of pumpkin is highly used and generates a high amount of waste, which can be used as a secondary product. This study aims to find the correlation between the extraction time, pH and temperature of Enzymatic assisted extraction of polysaccharides from *Cucurbita moschata* Duch.

Polysaccharides are essential macromolecules which almost exist in all living forms, and have important biological functions, they are getting more attention because they exhibit a wide range of biological and pharmacological activities, such as anti-tumor, immunomodulatory, antimicrobial, antioxidant, anticoagulant, antidiabetic, antiviral, and hypoglycemia activities, making them one of the most promising candidates in biomedical and pharmaceutical fields. Polysaccharides can be obtained from many different sources, such as plants, microorganisms, algae, and animals (Mohammed et al., 2021). Due to their physicochemical properties, they are susceptible to physical and chemical modifications leading to enhanced properties, which is the basic concept for their diverse applications in biomedical and pharmaceutical fields.

Although interest in polysaccharides is on the rise, along with confirmation of the efficiency of enzyme-assisted extraction, the effects of time and multivariate conditions of extraction parameters on the yield and quality of polysaccharides from *Cucurbita moschata* Duch remain to be systematically appraised. In fact, optimization of the process, which involves establishing relationships of each stage leading to maximal recovery and reproducibility, especially for its potential industrial use, is crucial. The present study deals with an investigation of the influence of time of extraction, pH and other important operational parameters on enzymatic release of polysaccharides, with the aim to develop optimal conditions for efficient and sustainable utilization of pumpkin residues as a valuable bioresource.

2. Materials and methods

2.1. Materials

The raw material used in this study was the pulp residue of *Cucurbita moschata* Duch (Collected from South-East Romania), with seeds removed. Pumpkin residue was washed, peeled, cut into small pieces, and dried at 50–60 °C until a constant weight was achieved. Then the dried pulp was grounded into a fine powder and stored in airtight containers at 4 °C until extraction. The enzyme used was pectinase, bought from Sigma-Aldrich.

2.2. Enzymatic assisted extraction of polysaccharides

EAE was performed using a commercially available enzyme such as pectinase. The extraction variables studied are extraction time (30, 45 and 60 minutes), temperature (25, 30, and 35 °C), pH (4, 5, 7) adjusted using a citrate-phosphate buffer (prepared from citric acid, Sigma-Aldrich, Germany, and disodium phosphate, Merck, Germany), enzyme concentration is always 2% w/v, and the solid–liquid ratio 1:10 g/mL. Experiments were performed in repetitions, N=6 times.

During the extraction process, pumpkin powder was suspended in buffer solution according to the selected solid–liquid ratio and the pH adjusted to the selected level. Afterwards, the enzyme was added at the selected concentration and incubation carried out at the selected temperature with constant stirring for the selected extraction time.

To terminate the enzymatic reaction, the mixture was heated at 95 °C for 10 minutes to inactivate the enzymes. Then, centrifuged at 5000 rpm for 10 minutes and collect the supernatant containing the polysaccharides. The centrifuge used is manufactured by Eppendorf, Germany.

2.3. Characterization

The total carbohydrate content was determined in the extracted polysaccharides by the phenol-sulfuric acid method (Masuko et.al, 2005). Further analyses should include determinations of molecular weight distribution by gel permeation chromatography, monosaccharide composition after hydrolysis followed by HPLC or GC, and structural characterization by FTIR or NMR spectroscopy.

The percentage of yield of PPs was determined using the equation below,

$$Yield = \frac{W_2}{W_1} \times 100 \text{ (1)}$$

Where, W_1 is the pumpkin powder weight and W_2 is the weight of recovered polysaccharides, characterized using the phenol-sulfuric acid method (Masuko et.al, 2005).

Following yield calculation, statistical analyses were conducted using Statistica (TIBCO, USA) to compare two-group and three-group models, considering time as the independent variable, before optimizing the experiment using response surface methodology.

2.4.Optimization by RSM

Response surface plots were presented in order to visualize the interaction among variables, and the optimum conditions for maximum yield of polysaccharides will be determined. Experiments will be carried out in triplicate, and data are represented as mean $\pm$ SD. Statistical analyses will be done using software such Statistica (TIBCO, USA).

3.Results and discussion

The following methodology yielded raw data, summarized in Table 1. In the table, the values marked 'a' were used for the two-group comparison, and the values marked 'b' were included in the three-group comparison.

This constitutes the basis for valid statistical analyses in accordance with the experimental design and enables appropriate comparisons between various experimental conditions. This approach allows for meaningful interpretation of the results, supporting robust conclusions about the effects observed under different experimental setups.

Table 1. Raw data for 2 groups and 3 groups

Conditions - extraction time (min)	Extraction yield (%)	Notes
30 min	12.4	a
30 min	11.9	a
30 min	11.5	a
30 min	12.5	a
30 min	13.1	a
30 min	12.9	a
45 min	18.1	a
45 min	18.3	a
45 min	17.9	a
45 min	17.5	a
45 min	17.8	a
45 min	18.2	a
60 min	22.5	b
60 min	24.3	b
60 min	25.2	b
60 min	21.8	b
60 min	22.8	b
60 min	23.7	b

Notes: Values used in the comparison between two groups are indicated by 'a', while values included in the comparison among three groups are indicated by 'b'.

3.1. Two-groups comparison

The Shapiro–Wilk test was performed using Statistica after input of the data for the two-groups comparison (Table 1. – Data marked with *a*). Obtained p-values were 0.8278 for 30-minute group and 0.8043 for 45-minute group; thus, both sets are normally distributed. To verify homogeneity of variances, the Brown–Forsythe test was used subsequently, which resulted in a p-value of 0.1959 confirming homogeneity. A t-test was performed and the p-value for the statistical comparison was less than 0.05.

The box-and-whisker plots with the means for each group are provided in Figure 1. The box-and-whisker plot for the Two- groups comparison indicates a marked increase in extraction yield with the increase in extraction time from 30 to 45 minutes. The average yield increased from 12.38% to 17.97% (45.15% increase). For both groups, data is normally distributed with small variability, no outliers detected. These visual trends align with the statistical tests confirming that the difference in extraction yield between the two extraction times is statistically significant.

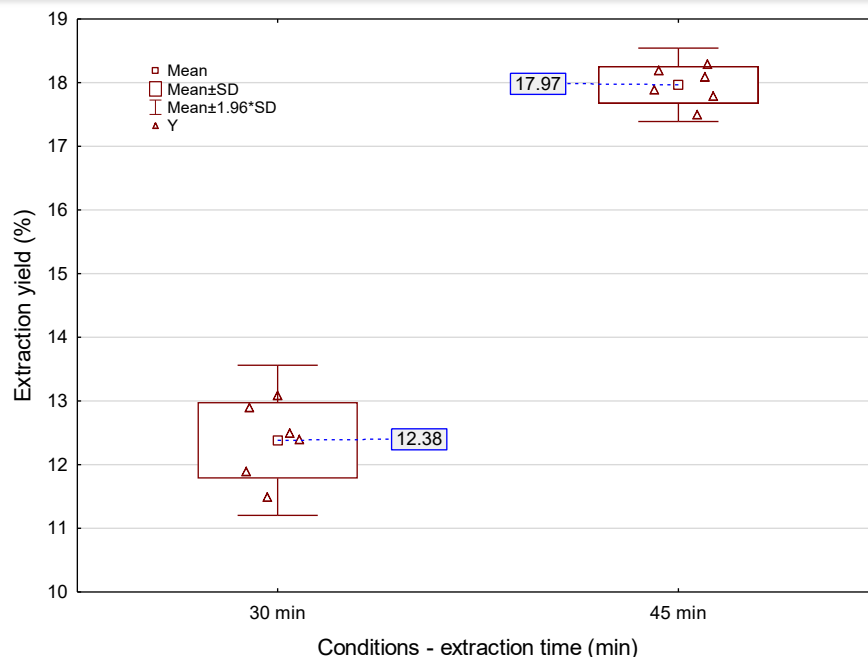


Figure 1.Box & Whisker Plots for Two-groups comparison

3.2.Three-groups comparison

The Shapiro–Wilk test was performed using Statistica from the data for the three-groups comparison (Table 1 – data marked with *a* and *b*). Obtained p-values were 0.8278 for 30-minute group and 0.8043 for 45-minute group and 0.9285 for 60-minute group; thus, all sets are normally distributed. To verify homogeneity of variances, the Brown–Forsythe test was used subsequently, which resulted in a p-value of 0.0130 denying homogeneity.

Thus the Kruskal – Wallis test was done, according to it, a statistically significant difference in extraction yield was observed only between the 30-minute and 60-minute groups, while the remaining comparisons did not meet the statistical significance threshold ($p < 0.05$).

The box & Whisker plots for Three-groups comparison are presented below in Figure 2. Extraction yield increases with time (30 min $\approx$ 12.38%, 45 min $\approx$ 17.97%, 60 min $\approx$ 23.38%) so a 45.15% increase from 30 mins to 45 mins and a 30.10% increase from 45 mins to 60 mins, with higher variability at longer times. Only the 30 min vs. 60 min difference is statistically significant ($p < 0.05$). For maximum yield, 60 min is optimal, while 45 min offers a balance between yield and consistency.

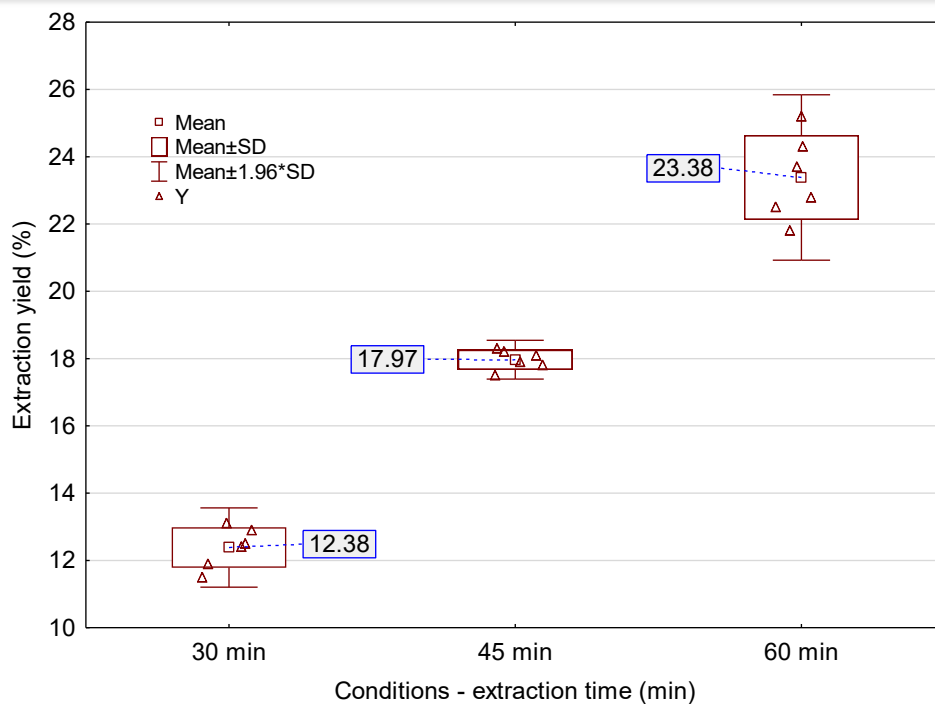


Figure 2.Box & Whisker Plots for Three-groups comparison

3.3.Experiment Optimization

In this study, extraction time, temperature, and pH were varied, and the corresponding results are summarized in the Table 2.

Table 2. Raw data for experiment optimization

Time (min)	Temperature (°C)	pH	Extraction yield (%)
30	25	5	8.90
60	25	5	20.83
30	35	5	16.32
60	35	5	25.33
30	30	4	13.32
60	30	4	19.56
30	30	7	15.38
60	30	7	22.68
45	25	4	10.52
45	35	4	14.49
45	35	7	26.30
45	25	7	18.99
45	30	4	13.73
45	30	4	12.20
45	30	4	13.04

The experimental results were investigated with a three-factor central composite design (CCD) in order to examine the effect of factors such as extraction time, temperature, and pH on the yield of the extract. The result from the Pareto chart revealed that the most significant positive impact on the extraction yield was from the extraction time, followed by the pH, and temperature. The significance of the models revealed that all three linear terms were significant (p -value < 0.05), but the quadratic terms, as well as interactive terms, were not significant.

Although some of the quadratic terms, as well as the interactions, were not significant ($p > 0.05$), they were still included in the equation because of hierarchy. The equations for the models are displayed below.

$$Y = 44.4797 - 0.03609(\text{Time}) + 0.00694(\text{Time})^2 - 2.88405(\text{Temperature}) + 0.05610(\text{Temperature})^2 - 2.13355(\text{pH}) + 0.11583(\text{pH})^2 - 0.00973(\text{Time} \times \text{Temperature}) - 0.00182(\text{Time} \times \text{pH}) + 0.10211(\text{Temperature} \times \text{pH}) \quad (2)$$

On this premise, that is, on the use of Pareto analysis, the factors that emerged most dominant in influencing the extraction yield as far as the impact is concerned are extraction time and pH., thus the RSM containing these factors is presented in Figure 3.

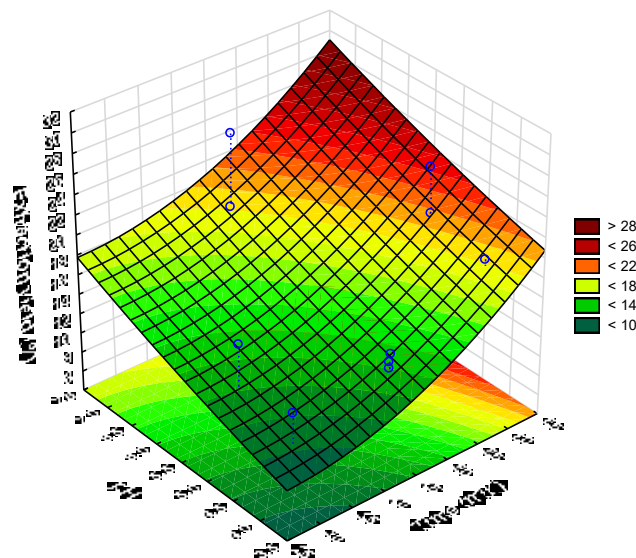


Figure 3. Response surface plot for experiment optimization

The 3-D graphical presentation of the results obtained from the RSM clearly indicates that the yield of the extraction is largely dependent on the factors of time and pH. The yield of the substance is higher with longer times of extraction, as well as with high pH. The highest yield is obtained when the extraction is carried out at a time of 75 to 80 minutes with a pH of 7 to 7.5. The curves on the graphical presentation indicate that there are slight nonlinear variations, but the most dominant factor is still time.

The factors to be used in the extraction process were optimized using the Desirability Function Technique, thereby giving the result for the optimum conditions necessary to reach the highest Extraction Yield (%). The profiles can be seen in Figure 4.

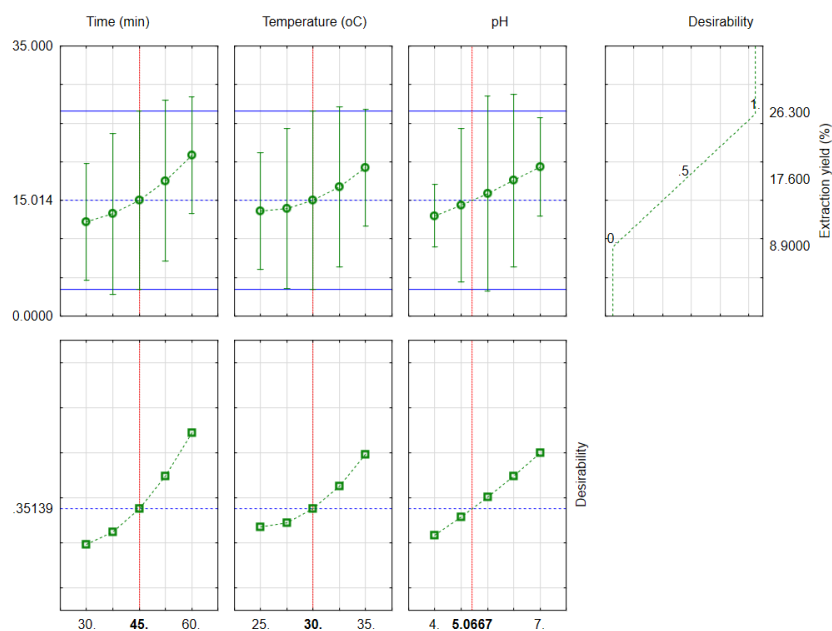


Figure 4. Profiles for predicted values and desirability

The analysis of the individual factors' profiles revealed that there is a positive relationship between the factors' values and the predicted value of Extraction Yield. The Desirability function increased with the increase in Time from 30 min to 60 min, Temperature from 25°C to 35°C, and with the increase in pH from 4.0 to 7.0.

The optimum point is the best compromise of factors where the multi-response desirability function is maximized, hence replicating the suggested conditions, that is, 45 min, 30°C, and pH 5.0667, as the conditions necessary to result in the highest extract yield.

4. Conclusions

Pectinase-assisted extraction turned out to be an efficient and "green" method for polysaccharides recovery from *Cucurbita moschata* Duch pulp residue. Among the process variables, extraction time was the most influential variable that affected yield, followed by pH, while temperature had a lesser effect within the range studied. Response surface methodology successfully modeled the extraction process, outlining optimal conditions at 45 minutes, 30 °C, and pH 5.07, as a good compromise between yield and efficiency of the process. Results validate enzyme-assisted extraction as valorization for pumpkin residues in a sustainable way and point out its potential. Future perspectives should study pilot or industrial scale application and reusability of the enzyme.

References

- Iqbal, A., Schulz, P., & Rizvi, S. S. H. (2021). Valorization of bioactive compounds in fruit pomace from agro-fruit industries: Present insights and future challenges. *Food Bioscience*, 44(Part A), 101384. <https://doi.org/10.1016/j.fbio.2021.101384>
- Umavathi, S., Keerthika, M., Gopinath, K., Kavitha, C., Romij Uddin, M., Alagumanian, S., & Balalakshmi, C. (2021). Optimization of aqueous-assisted extraction of polysaccharides from pumpkin (*Cucurbita moschata* Duch) and their biological activities. *Saudi Journal of Biological Sciences*, 28(12), 6692–6700. <https://doi.org/10.1016/j.sjbs.2021.07.034>
- Ezin, V., Gbemenou, U. H., & Ahanchede, A. (2022). Characterization of cultivated pumpkin (*Cucurbita moschata* Duchesne) landraces for genotypic variance, heritability and agro-morphological traits. *Saudi Journal of Biological Sciences*, 29(5), 3661–3674. <https://doi.org/10.1016/j.sjbs.2022.02.057>
- Mohammed, A. S. A., Naveed, M., & Jost, N. (2021). Polysaccharides: Classification, chemical properties, and future perspective applications in fields of pharmacology and biological medicine. *Journal of Polymers and the Environment*, 29(8), 2359–2371. <https://doi.org/10.1007/s10924-021-02052-2>
- Masuko, T., Minami, A., Iwasaki, N., Majima, T., Nishimura, S.-I., & Lee, Y. C. (2005). Carbohydrate analysis by a phenol–sulfuric acid method in microplate format. *Analytical Biochemistry*, 339(1), 69–72. <https://doi.org/10.1016/j.ab.2004.12.001>